

ExitPass MoPS Device Android Application Scope of Work

1. Project Title

ExitPass MoPS Device Android Application

2. Project Overview

This project covers the design and development of the Android application that will run on the MoPS device used in ExitPass continuity-mode operations. The application will support continuity-mode payment-related transaction capture, collection of required operational and transaction evidence, support for assigned incident and manual-operation workflows, and submission or handoff of captured records into the ExitPass continuity and reconciliation process.

The Android application will operate within the authority boundaries of ExitPass v1.2. A working Vendor PMS adapter and a mock Payment Orchestrator will be provided separately for integration and testing purposes. The student team will consume these provided interfaces but will not be responsible for developing them.

3. Project Objectives

The objectives of the project are as follows:

- Develop a working Android application for use on the MoPS device.
- Support continuity-mode transaction handling in a controlled and traceable manner.
- Capture the required continuity transaction details, operator inputs, and supporting evidence.
- Support the proper handoff of captured MoPS records into ExitPass continuity and reconciliation workflows.
- Ensure the mobile application remains within the authority, security, audit, and workflow boundaries defined by ExitPass v1.2.

4. In Scope

The student team's work shall include the following.

4.1 Android Application Development

- Design and develop the Android application for the MoPS device.
- Implement the required screens, forms, validation rules, and workflow navigation.
- Support user interaction flows for continuity-mode operations.

4.2 Continuity-Mode Payment-Related Transaction Capture

- Capture payment-related transaction details performed through the MoPS device during continuity mode.
- Capture required operational metadata, timestamps, identifiers, and supporting information relevant to each continuity transaction.
- Support continuity-specific transaction recording without treating the application as the source of final platform payment truth.

4.3 Operational Evidence Capture

- Capture required supporting evidence, notes, operator inputs, and operational context associated with continuity transactions.
- Support structured recording of relevant evidence needed for review, audit, and later reconciliation.

4.4 Incident and Exception Support

- Support logging or tagging of incidents, exceptions, operator notes, and related continuity context where required by the approved workflow.
- Support app-side workflows for handling operational exceptions relevant to the assigned scope.

4.5 Manual Operation Support

- Support device-side workflows related to assigned manual-operation handling where applicable.
- Support structured recording of actions that may later be reviewed or reconciled by backend services.

4.6 Integration with Provided Backend Interfaces

- Integrate the Android application with the working Vendor PMS adapter that will be provided.
- Integrate the Android application with the mock Payment Orchestrator that will be provided.
- Consume only the assigned APIs and backend services necessary for the MoPS device app workflow.

4.7 Reconciliation Handoff

- Submit, upload, or hand off captured MoPS transaction records into the ExitPass continuity and reconciliation flow through the prescribed backend interfaces.
- Support status visibility, acknowledgment, or review indicators if these are included in the approved application scope.

4.8 Basic Security and Access Control

- Implement login, session handling, and user-based or role-based access behavior as defined for the assigned application scope.
- Follow required security, audit, and traceability rules applicable to the Android application.

4.9 Testing and Documentation

- Prepare functional documentation, screen flow documentation, and technical notes for the Android application.
- Perform unit, functional, and integration testing appropriate to the student project scope.
- Deliver test cases, test results, known limitations, and usage notes.

5. Out of Scope

The following are explicitly excluded from the student team's scope of work.

- Ticket issuance
- Entry processing
- New parking session creation
- Development of the Vendor PMS adapter
- Development of the Payment Orchestrator
- Real payment-provider integration
- Provider callback verification
- Platform payment finalization
- Standard online ExitPass payment flows outside continuity mode
- Exit authorization issuance
- Live gate-open control or barrier control
- Production deployment to live operating sites
- Production-grade infrastructure hardening beyond the student project requirement

6. Assumptions

The following assumptions apply to this scope of work.

- A working Vendor PMS adapter will be provided for the student team to integrate with.
- A mock Payment Orchestrator will be provided for the student team to integrate with.
- The Android application will be developed as a bounded subsystem within ExitPass and not as an independent authority system.
- Backend services required for app integration will be made available in a development or test environment appropriate for the project.

7. Implementation Boundaries

The Android application must comply with the following ExitPass v1.2 boundaries.

- The application is a continuity-mode operational tool, not the platform authority for payment finality.
- Vendor PMS remains authoritative for raw parking-session lifecycle and tariff computation.
- Central PMS remains authoritative for payment-linked control state and reconciliation acceptance.
- MoPS transactions and manual operational records remain continuity and exception evidence until processed through reconciliation, and must not silently become normal payment-attempt records.
- The provided mock Payment Orchestrator may be used for application workflow completion and testing, but the Android app must not assume responsibility for orchestration or platform finality.

8. Expected Deliverables

The student team is expected to deliver the following.

- Android application source code for the MoPS device
- UI or screen flow documentation
- Functional scope summary and feature list
- Technical design notes relevant to the mobile application
- API integration notes for the application's assigned backend interfaces
- Test cases and test results
- User guide or operator guide for the Android application
- Final presentation or demonstration of the working prototype

9. Recommended Scope Statement for Handoff

The student team will design and develop the Android application for the ExitPass MoPS device. The application's scope is limited to continuity-mode payment-related transaction capture, collection of required operational evidence, support for assigned incident and manual-operation workflows, and integration with a provided Vendor PMS adapter and a

provided mock Payment Orchestrator for testing and workflow completion. The scope includes handoff of MoPS records into ExitPass continuity and reconciliation processes. The scope does not include ticket issuance, entry processing, Vendor PMS adapter development, Payment Orchestrator development, real payment-provider integration, platform payment finalization, standard online payment flows, exit authorization logic, or live gate control.